

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-002 - Nahorkatiya-002

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000014
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2001-04-13 / 2001-08-30
Measured / TVD	2606 m / 2591 m
Trajectory	Deviated; inc 24.0 deg; az 345.1 deg
Formation Interval	1479-2356 m
Lithology	Sandstone-siltstone sequence
Pore Pressure Reference	2846 psi
Fracture Gradient	14.84 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	High (60.8/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	651	Surface
9 5/8 in casing	1511	Intermediate
7 in liner/casing	2293	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2001-05-09 622 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2001-05-15 743 m	Pressure Spike	High	Circulating pressure climbed without equivalent torque change. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2001-06-20 1392 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2001-07-05 1644 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2001-07-07 1681 m	Cementing Issue	High	Cement returns were below planned percentage. Mitigation: Control displacement rate and reconcile pumped/returned volumes. Lesson: Plan conservative displacement rate in weak formations.
2001-08-21 2445 m	Mud Loss	High	Active pit volume declined during stable pump rate. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2001-08-24 2508 m	Kick	High	Unexpected background gas rise preceded influx. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Maintain mud-weight window using offset pressure evidence.

Completion Summary

The synthetic well reached 2606 m MD in the Tipam interval. The NWIS knowledge record links 20 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.